OCT
13

Coding challenge_Tinkecad
Ishita Bhattacharjee
Classwork 16/18 Students Tuesday at 11:35 AM
IB MYP Design - Ms Ishita (MYP2) B

Task Details

Tinkercad Arduino Circuit Challenges

Use **Tinkercad Circuits** to design, build, simulate, and test the following Arduino-based challenges. Your circuit must function correctly in the simulation and your code should be clearly structured and commented.

Challenge 1: Traffic Light System

Create a traffic light system using **three LEDs (red, yellow and green)**.

Program the Arduino so that:

- 🔴 **Red LED** stays ON for **5 seconds**
- 🟡 **Yellow LED** stays ON for **1 second**
- 🟢 **Green LED** stays ON for **7 seconds**
- The sequence then repeats continuously.

You should use appropriate resistors, connect the components correctly, and write an Arduino program to control the timing.

Challenge 2: Automatic night-light system

Using **Tinkercad Circuits**, design and simulate an **automatic night-light system** using an Arduino Uno, an LDR, and an LED.

Your system should be able to **detect the surrounding light level and respond automatically**. Program the Arduino so that the LED turns **ON when the surroundings are dark** and turns **OFF when the surroundings are bright**.

You must:

- Connect the LDR to the Arduino to detect changes in light intensity.
- Connect an LED with an appropriate resistor.
- Write an Arduino program that reads the LDR value and makes a decision based on the light level.
- Test your circuit by changing the light level in the Tinkercad simulation.
- Identify and set an appropriate **threshold value** that determines when the LED should turn ON or OFF.